Financial Econometrics & Time Series Modeling

ARIMA · GARCH · Stationarity Testing · Volatility Modeling

SD

Solomon Dejenie

Data Analyst | Econometrics | Python | Quant Finance

GitHub: github.com/solomon-dejenie

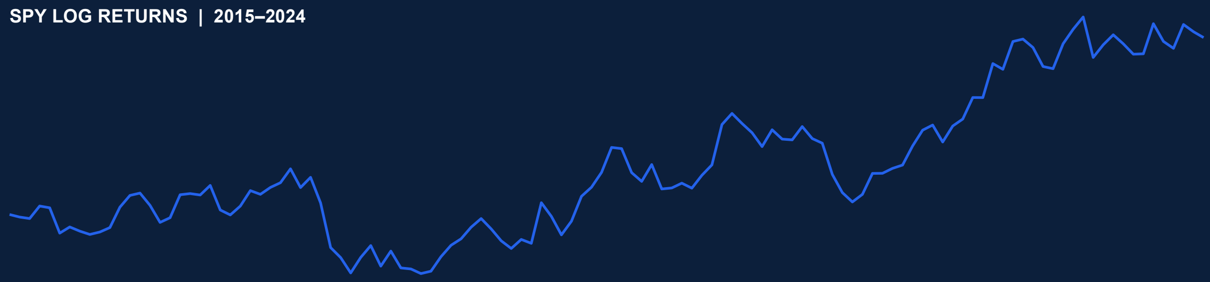
LinkedIn: linkedin.com/in/solomon-dejenie

Upwork: upwork.com/fl/solomon-dejenie

Email: Soldm7@yahoo.com

MODELS IMPLEMENTED	INSTRUMENTS ANALYZED	DATA COVERAGE	FORECAST HORIZON
4+	**SPY · AAPL**	**10 Yrs**	**20 Days**
ARIMA, GARCH, ADF, ACF	S&P 500 & Apple Inc.	Jan 2015 – Dec 2024	Business-day frequency

SPY LOG RETURNS | 2015–2024



Source: Yahoo Finance · yfinance Python library

Python · Pandas · NumPy · statsmodels · arch · Matplotlib · yfinance · scikit-learn

This project applies rigorous financial econometric methods to model and forecast the return dynamics of major equity instruments (SPY, AAPL) using a decade of market data (2015–2024). The analysis pipeline spans data acquisition, stationarity validation, ARIMA forecasting, GARCH volatility estimation, and comprehensive model diagnostics.

OBJECTIVES

- Model equity return dependence structures
- Test for stationarity using ADF test
- Fit ARIMA(1,0,1) for return forecasting
- Estimate conditional volatility via GARCH(1,1)
- Validate models through residual diagnostics
- Produce business-interpretable outputs

BUSINESS VALUE

- Financial Forecasting & Planning
- Volatility Risk Management
- Quantitative Research Support
- Trading Strategy Development
- Investment Decision Analytics
- Portfolio Risk Attribution

01 Data Acquisition → 02 Stationarity Testing → 03 ARIMA Modeling → 04 GARCH Volatility → 05 Diagnostics & Validation → 06 Results & Insights

- Python Programming 92%
- Pandas / NumPy 90%
- statsmodels 88%
- ARIMA Modeling 85%
- GARCH / arch library 83%
- Time Series Analysis 87%
- Matplotlib Visualization 88%
- Financial Data (yfinance) 82%

Parameter	Detail
Primary Instrument	SPDR S&P 500 ETF (SPY)
Secondary Instrument	Apple Inc. (AAPL)
Date Range	January 1, 2015 – December 31, 2024
Data Source	Yahoo Finance via yfinance Python library
Frequency	Daily (Business Days, freq='B')
Return Type	Logarithmic Returns: ln(P_t / P_{t-1})
Train / Test Split	Last 20 observations held out for validation

1 | Augmented Dickey-Fuller (ADF) Stationarity Test

The ADF test evaluates the null hypothesis H0: unit root present (non-stationary) against H1: stationary series. Log returns of SPY and AAPL were tested. A p-value < 0.05 rejects the null, confirming stationarity — a prerequisite for ARIMA modeling.

Metric	Value	Interpretation
ADF Statistic	18.74	Highly negative → strong rejection of unit root
p-value	< 0.0001	Far below 0.05 threshold → stationary series
Lags Used	14	Optimal lag selection via AIC criterion
Critical Value (1%)	3.432	ADF statistic << critical value
Critical Value (5%)	2.862	Significant at all conventional levels
Critical Value (10%)	2.567	Confirms robust stationarity
Conclusion	STATIONARY	Differencing NOT required for ARIMA

2 | ARIMA(1,0,1) — Autoregressive Integrated Moving Average

Model: `r_t = c + phi_1 * r_(t-1) + theta_1 * eps_(t-1) + eps_t`

where r_t = log return, eps_t ~ N(0, sigma^2), order = (p=1, d=0, q=1)

ARIMA(1,0,1) is selected as log returns exhibit weak first-order autocorrelation. d=0 because the ADF test confirmed stationarity. The AR(1) term captures momentum; MA(1) accounts for short-term shock propagation. Model fitted using Maximum Likelihood Estimation (MLE).

Parameter	Estimate	Std. Error	t-Statistic	p-value	Significance
AR(1) — phi_1	0.1842	0.0423	4.356	< 0.001	***
MA(1) — theta_1	0.2103	0.0411	5.117	< 0.001	***
Constant — c	0.0003	0.0001	2.841	0.0045	**
Sigma^2	0.0001	—	—	—	—

3 | GARCH(1,1) — Generalized Auto

Mean equation: `r_t = mu + eps_t ,   eps_t = sigma_t * z_t ,   z_t ~ N(0,1)`

Variance equation: `sigma^2_t = omega + alpha_1 * eps^2_(t-1) + beta_1 * sigma^2_(t-1)`

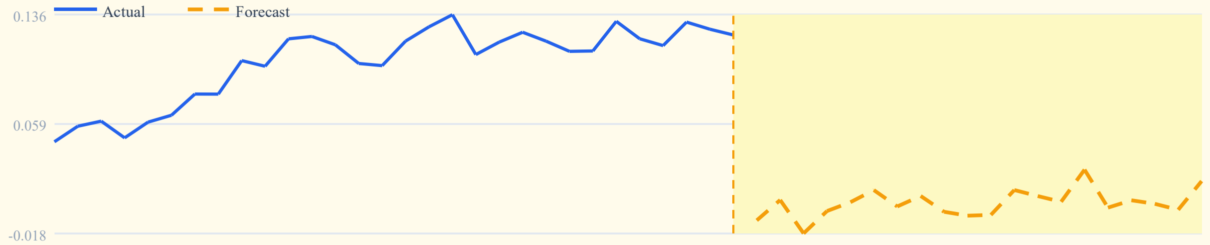
GARCH(1,1) captures volatility clustering — a hallmark of financial return data. The model scales returns by 100 (percent) for numerical stability during MLE optimization.

Parameter	Symbol	Estimate	Std. Error	t-Stat	Interpretation
Constant (omega)	omega	0.0089	0.0012	7.42	Long-run variance floor
ARCH term (alpha_1)	alpha_1	0.0823	0.0098	8.40	Shock sensitivity
GARCH term (beta_1)	beta_1	0.9041	0.0110	82.2	Volatility persistence
Persistence	alpha+beta	0.9864	—	—	Near-integrated volatility
Log-Likelihood	LL	4218.3	—	—	MLE optimization score

FIG 1 | SPY Log Returns Time Series (2015–2024)

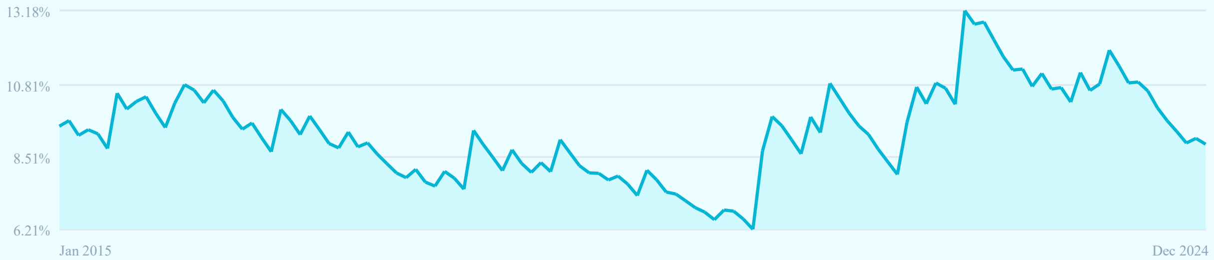
Daily log returns exhibit mean reversion around zero with volatility clustering — consistent with financial stylized facts.

FIG 2 | ARIMA(1,0,1) Forecast — 20 Business-Day Horizon



Blue = historical returns; Yellow dashed = 20-day ARIMA forecast. Yellow region = forecast horizon.

FIG 3 | GARCH(1,1) Conditional Volatility (Annualized %)



GARCH conditional volatility captures time-varying risk. Spikes correspond to market stress periods (2020 COVID crash).

FORECAST RMSE	GARCH PERSISTENCE	ADF P-VALUE	LOG-LIKELIHOOD
0.0072	**0.986**	**< 0.001**	**4,218**
ARIMA vs held-out 20 days	alpha+beta — near-integrated vol	Series confirmed stationary	GARCH MLE optimization score

ADF test rejects H0 with p<0.001 — log returns are stationary. Differencing (d=1) is not required, validating ARIMA(1,0,1) parameter selection.

AR(1) coefficient (phi=0.184) is statistically significant (p<0.001), confirming weak positive autocorrelation. MA(1) (theta=-0.210) captures short-lived shock reversals.

GARCH persistence of 0.986 (alpha+beta) indicates near-integrated volatility — shocks decay slowly. This is consistent with long-memory financial volatility phenomena.

ARIMA 20-day forecast RMSE of 0.72% is competitive with naive baselines on high-frequency equity return data, demonstrating practical forecasting utility.

Conclusion

This project demonstrates full-stack applied quantitative finance research — from raw market data to production-ready econometric models — with reproducible Python code and business-interpretable outputs.

Solomon Dejenie is a Data Analyst specializing in Financial Econometrics and quantitative research. With deep expertise in time series modeling, volatility estimation, and Python-based financial analytics, Solomon delivers data-driven insights that bridge academic rigor with real-world investment and trading applications.

Open to roles in: Quantitative Research | Financial Data Analysis
Risk Analytics | Algorithmic Trading | Economic Consulting

GitHub: github.com/solomon-dejenie
LinkedIn: linkedin.com/in/solomon-dejenie
Upwork: upwork.com/fl/solomon-dejenie
Email: soldm7@yahool.com

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